

Adaptation Without Strict Execution Contraction

A proposed theory section for frozen VLA policies

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This draft synthesizes the evidence at repository commit `b540d52`. Its mathematical statements are conditional; the experiments do not certify their physical bounds. The supporting note records the empirical evidence, counterexamples, and a corrected telemetry audit. This standalone draft is not inserted into the submission manuscript.

1 Observation, correction, and task execution

A frozen policy produces a nominal command a_k , and the adapter sends $u_k = a_k - c_k$. The healthy predictor must use the known command u_k . At a specified state and horizon, distinguish the residual observation r_k from the physical discrepancy v_k remaining after correction:

$$r_k = H_k \theta_k + \varepsilon_k, \quad v_k = F_k \theta_k - D_k c_k. \quad (1)$$

The physical disturbance map F_k , observation map H_k , and available-input map D_k need not coincide. For any nominal allocation K_k ,

$$v_k = (F_k - D_k K_k) \theta_k + D_k K_k (\theta_k - \hat{\theta}_k) + D_k (K_k \hat{\theta}_k - c_k). \quad (2)$$

The identity separates error in the cancellation map, estimation error, and error in allocating and realizing the correction. The first term concerns the chosen allocation; it need not be irreducible.

In the noiseless unconstrained linear case, a linear rule $c = Lr$ cancels every disturbance exactly iff

$$\ker H \subseteq \ker F, \quad \text{range } F \subseteq \text{range } D. \quad (3)$$

Indeed, the first inclusion makes $T(H\theta) = F\theta$ well-defined and the second permits lifting T through D . Constraints additionally require pointwise feasibility $F\theta \in DC$. Existence of one feasible linear rule over a disturbance set further requires $LH\theta \in \mathcal{C}$ throughout that set. Accurate observation therefore does not imply feasible cancellation, and a reduced fitted objective does not by itself imply physical improvement.

2 Two implemented estimator recursions

For command-equivalent additive faults, write

$$z_k = M^{-1} r_k - b = f_k + w_k, \quad e_k = \hat{f}_k - f_k, \quad E_k = \|e_k\|, \quad \nu_k = \|f_{k+1} - f_k\|.$$

Assume $f_k \in \mathcal{C}$, with $\mathcal{C}$ closed and convex, Euclidean projection $\Pi_{\mathcal{C}}$, and $\|w_k\| \leq \epsilon_k$. The latter must be independently bounded to obtain a predictive guarantee.

Proposition 1 (Legacy and innovation bounds). *For $0 < \gamma \leq 1$ and $0 \leq s_k \leq 1$, the legacy update*

$$\hat{f}_{k+1} = \Pi_C[(1 - \gamma)\hat{f}_k + \gamma s_k z_k]$$

satisfies

$$E_{k+1} \leq (1 - \gamma)E_k + \gamma s_k \epsilon_k + \gamma(1 - s_k)\|f_k\| + \nu_k. \quad (4)$$

For $0 \leq \alpha_k \leq 1$, the innovation update

$$\hat{f}_{k+1} = \Pi_C[\hat{f}_k + \alpha_k(z_k - \hat{f}_k)]$$

satisfies

$$E_{k+1} \leq (1 - \alpha_k)E_k + \alpha_k \epsilon_k + \nu_k. \quad (5)$$

Proof. Before projection, subtract f_k . The respective expressions are $(1 - \gamma)e_k + \gamma s_k w_k - \gamma(1 - s_k)f_k$ and $(1 - \alpha_k)e_k + \alpha_k w_k$. Nonexpansiveness of projection relative to f_k , the triangle inequality, and the change from f_k to f_{k+1} give the results. $\square$

The legacy zero-target deadzone is $s_k = 0$; an explicit hold branch is a different update. Innovation uses $\alpha_k = \gamma s_k$, with its normalizer evaluated on the innovation. Persistent w_k biases both estimators. Only legacy has the additional attenuation forcing in (4). For constant positive α , the innovation steady bound is $\epsilon + \nu/\alpha$. Holding at T instead gives

$$\|\hat{f}_T - f_{T+j}\| \leq E_T + \sum_{\ell=T}^{T+j-1} \nu_\ell. \quad (6)$$

This also describes snapshot age. Holding removes subsequent estimator updates; it retains calibration error and fault-drift sensitivity.

3 Finite-horizon incremental storage

Augment the physical deviation with the controller/decoder memory, predictor history, estimator state, and relevant policy/chunk state. Let the resulting deviation be Z_k . Relative to a specified nominal trajectory, suppose the incremental dynamics on a declared domain satisfy

$$Z_{k+1} = \mathcal{A}_k Z_k + \mathcal{B}_k d_k + \rho_k. \quad (7)$$

The signed transition includes estimator-execution feedback; d_k, ρ_k include disturbance drift, unmatched effects, nonlinear remainder, and unmodeled inputs. A healthy output-prediction fit alone does not establish (7) or uniform remainder bounds.

Theorem 1 (Finite-horizon storage bound). *Fix a finite horizon N . Choose a positive-definite metric P_k at each time $k = 0, \dots, N$. Define*

$$V_k = Z_k^\top P_k Z_k, \quad a_k = \left\| P_{k+1}^{1/2} \mathcal{A}_k P_k^{-1/2} \right\|_2,$$

and suppose $\|\mathcal{B}_k d_k + \rho_k\|_{P_{k+1}} \leq b_k$. If $R_0 \geq \sqrt{V_0}$, then

$$\sqrt{V_k} \leq R_k, \quad R_{k+1} = a_k R_k + b_k, \quad (8)$$

$$R_n = \left(\prod_{j=0}^{n-1} a_j \right) R_0 + \sum_{k=0}^{n-1} \left(\prod_{j=k+1}^{n-1} a_j \right) b_k. \quad (9)$$

Proof. The induced-norm and triangle inequalities give $\sqrt{V_{k+1}} \leq a_k \sqrt{V_k} + b_k$. Induction yields both claims. $\square$

The theorem permits $a_k = 1$ and finite periods with $a_k > 1$. It establishes a finite-horizon bound, not asymptotic stability. Report metric coercivity and physical-output conversion: shrinking P_k does not make a growing physical deviation harmless. For local assumptions, verify the tube stays within their domain. Keeping signed matrix transition products can be substantially tighter than products of scalar norms.

Corollary 1 (Strict contraction as a special case). *If $a_k \leq a < 1$ and $b_k \leq b$, then $R_n \leq a^n R_0 + b(1 - a^n)/(1 - a)$.*

Alternatively, a separately justified physical inequality $X_{k+1} \leq \lambda_k X_k + L_k \|v_k\| + \eta_k$ fits the theorem with (2). Its input is the remaining physical discrepancy v_k , not the predictor residual r_k . Contact/controller resets require additional jump inequalities and domain conditions; a fitted free-space metric does not cover them automatically.

4 Execution memory explains why identification and repair differ

Consider an exact scalar example, $0 < \alpha \leq 1$:

$$p_{k+1} = \lambda p_k + g(f - \hat{f}_k), \quad \hat{f}_{k+1} = (1 - \alpha)\hat{f}_k + \alpha f, \quad p_0 = \hat{f}_0 = 0.$$

The remaining fault is $f(1 - \alpha)^k$. For $\lambda \neq 1 - \alpha$,

$$p_n = gf \frac{\lambda^n - (1 - \alpha)^n}{\lambda - (1 - \alpha)}; \quad (10)$$

the coincident-root value is $gf n \lambda^{n-1}$. For a stable servo, $0 \leq \lambda < 1$, the displacement decays. For an integrator, $\lambda = 1$,

$$p_n = \frac{gf}{\alpha} [1 - (1 - \alpha)^n] \rightarrow \frac{gf}{\alpha}. \quad (11)$$

Thus an unbiased observer stops further drift but need not erase transient displacement. With legacy attenuation $0 \leq s < 1$ and $0 < \gamma \leq 1$,

$$p_n = gf \left[(1 - s)n + \frac{s}{\gamma} (1 - (1 - \gamma)^n) \right]. \quad (12)$$

Persistent bias and transient cost enter differently. For an ideal integrator held for h steps, (6) bounds the subsequent displacement contribution by $|g|[hE_T + \nu h(h - 1)/2]$. Identification followed by a task reset can remove the earlier displacement; merely pausing an estimate during a disrupted trajectory does not. These are constructed examples, not fitted explanations of individual runs.

5 Joint Lyapunov stability without physical subsystem contraction

Consider a normalized neutral channel, constant fault, exact observation $z_k = f$, and no projection. Apply the composite update $\hat{f}_{k+1} = \hat{f}_k + \alpha(z_k - \hat{f}_k) + \kappa e_{k+1}$. With $\delta_k = \hat{f}_k - f$, the error system is

$$e_{k+1} = e_k - \delta_k, \quad \delta_{k+1} = (1 - \alpha)\delta_k + \kappa e_{k+1}. \quad (13)$$

Proposition 2 (Stabilization of a neutral execution mode). *The joint matrix*

$$\mathcal{A} = \begin{bmatrix} 1 & -1 \\ \kappa & 1 - \alpha - \kappa \end{bmatrix}$$

is Schur stable iff $0 < \alpha < 2$ and $0 < \kappa < 4 - 2\alpha$.

Proof. $\det \mathcal{A} = 1 - \alpha$ and $\text{tr } \mathcal{A} = 2 - \alpha - \kappa$. The strict Jury inequalities reduce to $\alpha > 0$, $\kappa > 0$, and $4 - 2\alpha - \kappa > 0$. $\square$

For every $Q \succ 0$ there consequently exists $P \succ 0$ satisfying $\mathcal{A}^\top P \mathcal{A} - P = -Q$. For $Z_{k+1} = \mathcal{A}Z_k + w_k$ and $q = \lambda_{\min}(Q)$,

$$V_{k+1} - V_k \leq -\frac{q}{2} \|Z_k\|^2 + \left(\|P\|_2 + \frac{2\|\mathcal{A}^\top P\|_2^2}{q} \right) \|w_k\|^2. \quad (14)$$

This follows by expanding the quadratic and applying Young's inequality. It gives a joint Lyapunov/ISS bound even though the physical subsystem alone is neutral. At $\kappa = 0$ the neutral eigenvalue remains; excessive tracking gain destabilizes the system.

This is a conditional design result. The actual reference, covariance, input map, sampling, clipping, and contact behavior must be included before applying it to a robot. Increment-only tracking may not observe a neutral pose mode. Stable individual frozen-gain matrices do not establish stability under arbitrary gain switching.

6 Conditional task transfer and empirical interpretation

Same-command healthy physical execution differs from the healthy closed-loop policy counterfactual. A frozen policy generally emits different commands after a disturbance. Include that feedback in Z_k , or bound the command difference separately; a predictor residual is neither trajectory.

Suppose a successful healthy trajectory has robustness radius $\mu > 0$ in the weighted supremum metric $d_{\text{task}} = \max_{k \leq N} \|W_k(x_k - x_k^0)\|$. If $x_k - x_k^0 = C_k Z_k$ and

$$\max_{k \leq N} \|W_k C_k P_k^{-1/2}\|_2 R_k < \mu, \quad (15)$$

with compatible domain and hybrid assumptions, the corrected trajectory also succeeds. This is a consequence of the assumed success neighborhood, not a margin identified by the present experiments. Static-offset sweeps do not characterize arbitrary time-varying trajectory errors.

The evidence motivates this distinction: same-command Panda perturbations do not establish strict contraction, while repair remains observed; a matched-estimate ALOHA innovation observer repairs without holding; improved ARX prediction does not remove rotation-y observation bias; and the pinned static-observer comparison does not resolve a FIR-memory benefit. The finite-horizon theorem accommodates these outcomes without asserting they certify its constants. The authority and joint Lyapunov analyses also permit harmful correction and destabilizing tracking feedback.

The theoretical tools are standard. The proposed contribution is their faithful connection to the implemented VLA interface and explicit empirical tests of the required assumptions, not universal adaptive recovery.

References

- [1] F. Forni and R. Sepulchre. A differential Lyapunov framework for contraction analysis. *IEEE Transactions on Automatic Control*, 2014. <https://arxiv.org/abs/1208.2943>.
- [2] F. Forni and R. Sepulchre. On differentially dissipative dynamical systems, 2013. <https://arxiv.org/abs/1305.3456>.
- [3] R. Geiselhart and F. Wirth. Relaxed ISS Small-Gain Theorems for Discrete-Time Systems. <https://arxiv.org/abs/1406.3224>.
- [4] Y. Cheng et al. Improving the Robustness of Reinforcement Learning Policies with $\mathcal{L}_1$ Adaptive Control. *IEEE Robotics and Automation Letters*, 2022. <https://arxiv.org/abs/2112.01953>.
- [5] J.-J. E. Slotine and W. Li. Composite adaptive control of robot manipulators. *Automatica*, 1989. [https://doi.org/10.1016/0005-1098\(89\)90094-0](https://doi.org/10.1016/0005-1098(89)90094-0).

The nonstrict/incremental and input-supply interpretations follow [1, 2]; finite-step interconnection analysis is discussed in [3]. Frozen-policy augmentation and composite adaptation have established antecedents [4, 5]. These works do not automatically certify the particular discrete controller studied here.